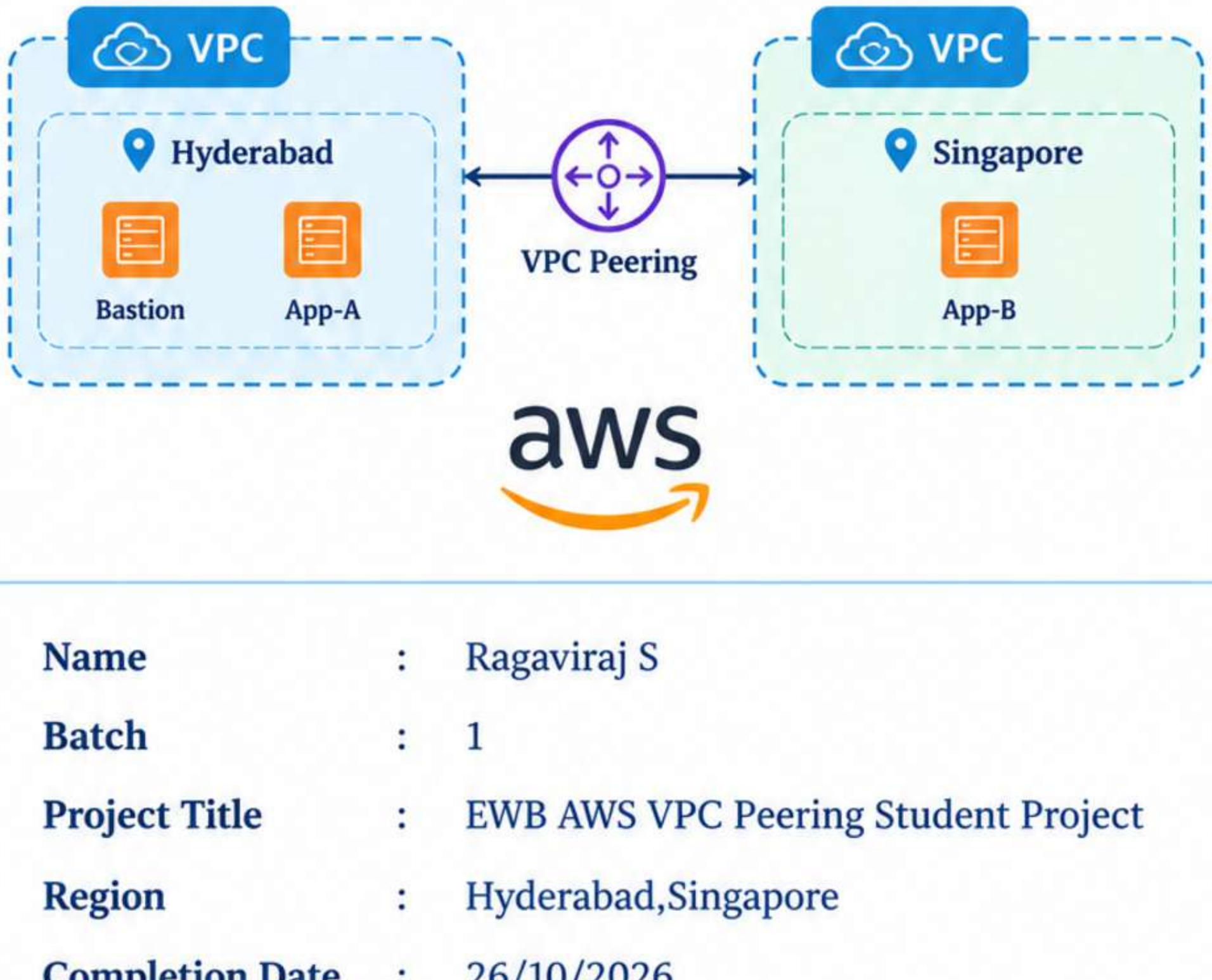
EWB

EWB CLOUD LABS

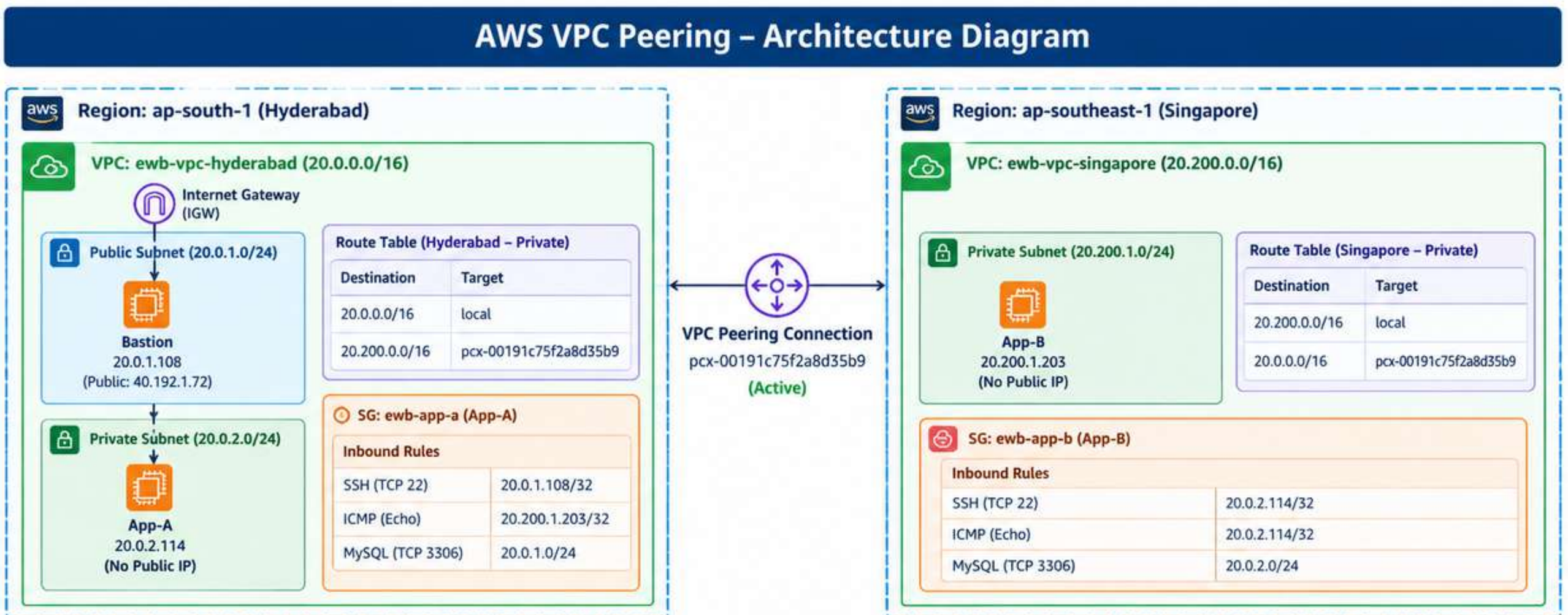
Student Project Report

EWB AWS VPC Peering Student Project



Name	:	Ragaviraj S
Batch	:	1
Project Title	:	EWB AWS VPC Peering Student Project
Region	:	Hyderabad,Singapore
Completion Date	:	26/10/2026

AWS VPC Peering – Architecture Diagram



1. VPC Details

Resource / Name	VPC ID	CIDR Block	Region
ewb-vpc-hyderabad	vpc-0f0cf477a66421274	20.0.0.0/16	ap-south-1
ewb-vpc-singapore	vpc-0869c24a0864f20bf	20.200.0.0/16	ap-southeast-1

2. Subnet Details

Resource / Name	Subnet ID	CIDR Block	VPC
Bastion subnet	subnet-0a392df69de769963	20.0.1.0/24	Hyderabad
App-A private subnet	subnet-0a392df69de769963	20.0.2.0/24	Hyderabad
App-B private subnet	subnet-0b63e3f15c871eecb	20.200.1.0/24	Singapore

3. EC2 Instances

Resource / Name	Instance ID	Private IP	Public IP	VPC
Bastion	i-0bo[illegible]c1a108	20.0.1.108	40.192.1.72	Hyderabad
App-A	i-0do[illegible]c2a114	20.0.2.114	–	Hyderabad
App-B	i-0f1dbd41a453424d5	20.200.1.203	–	Singapore

4. Security Groups

Resource / Name	Inbound Rules	VPC
ewb-app-a (App-A)	SSH: 20.0.1.108/32 ICMP: 20.200.1.203/32 MySQL: 20.0.1.0/24	Hyderabad
ewb-app-b (App-B)	SSH: 20.0.2.114/32 ICMP: 20.0.2.114/32 MySQL: 20.0.2.0/24	Singapore

5. VPC Peering

Peering Connection ID	Status	Requester VPC	Accepter VPC	CIDR Blocks
pcx-00191c75f2a8d35b9	Active	vpc-0f0cf477a66421274 (20.0.0.0/16)	vpc-0869c24a0864f20bf (20.200.0.0/16)	20.0.0.0/16 ↔ 20.200.0.0/16

6. Legend

- VPC
- Public Subnet
- Private Subnet
- EC2 Instance
- Internet Gateway (IGW)
- VPC Peering Connection
- Traffic Flow
- Route Association

AWS VPC Peering – Resource and IP Details

1 Resources and IP Details

Resources / Name	VPC or Subnet	CIDR / IP	Purpose
ewb-vpc-hyderabad	Hyderabad VPC	20.0.0.0/16	Hosts Bastion and App-A resources
ewb-vpc-singapore	Singapore VPC	20.200.0.0/16	Hosts App-B resources
ewb-bast	Hyderabad VPC	Private IP: 20.0.1.108 Public IP: 40.192.1.72	Bastion host for SSH access to private instances
Bastion subnet	Hyderabad VPC	20.0.1.0/24	Deploys Bastion host
ewb-app-a	Hyderabad VPC	Private IP: 20.0.2.114	App-A instance; connectivity test with App-B
App-A private subnet	Hyderabad VPC	20.0.2.0/24	Hosts App-A instance
ewb-app-b	Singapore VPC	Private IP: 20.200.1.203	App-B instance; cross-region connectivity test
App-B private subnet	Singapore VPC	20.200.1.0/24	Hosts App-B instance
pcx-00191c75f2a8d35b9	Hyderabad ↔ Singapore VPCs	20.0.0.0/16 ↔ 20.200.0.0/16	Enables private communication between both VPCs
ewb-app-a Security Group	Hyderabad VPC	SSH: 20.0.1.108/32 ICMP: 20.200.1.203/32 MySQL: 20.0.1.0/24	Controls inbound traffic to App-A
ewb-app-b Security Group	Singapore VPC	SSH: 20.0.2.114/32 ICMP: 20.0.2.114/32 MySQL: 20.0.2.0/24	Controls inbound traffic to App-B

2 Route Table Details

Resource / Name	VPC or Subnet	CIDR / IP	Target	Purpose
Hyderabad private route table	Hyderabad VPC	20.0.0.0/16	Local	Same VPC communication
Hyderabad private route table	Hyderabad VPC	20.200.0.0/16	pcx-00191c75f2a8d35b9	Singapore VPC traffic route
Singapore private route table	Singapore VPC	20.200.0.0/16	Local	Same VPC communication
Singapore private route table	Singapore VPC	20.0.0.0/16	pcx-00191c75f2a8d35b9	Hyderabad VPC traffic route

3 VPC Peering Details

Peering Connection ID	Status	Requester VPC	Accepter VPC	CIDR Blocks
pcx-00191c75f2a8d35b9	**Active**	20.0.0.0/16 (Hyderabad)	20.200.0.0/16 (Singapore)	20.0.0.0/16 ↔ 20.200.0.0/16

vpc-0121900efe558e38d / hyderabadvpc

Actions

Details Info

VPC ID
vpc-0121900efe558e38d

State
Available

Block Public Access
Off

DNS hostnames
Disabled

DNS resolution
Enabled

Tenancy
default

DHCP option set
dopt-07ada3796956400fc

Main route table
rtb-03c57c4d7b8fe81a9

Main network ACL
acl-0162ee07865da9c49

Default VPC
No

IPv4 CIDR
20.0.0.0/16

IPv6 pool
–

IPv6 CIDR
–

Network Address Usage metrics
Disabled

Route 53 Resolver DNS Firewall rule groups
–

Owner ID
112334179662

Encryption control ID
–

Encryption control mode
–

vpc-0a5283e29e2d2a530 / sing-vpc

Actions

Details Info

VPC ID
vpc-0a5283e29e2d2a530

State
Available

Block Public Access
Off

DNS hostnames
Disabled

DNS resolution
Enabled

Tenancy
default

DHCP option set
dopt-067debf4ef3795186

Main route table
rtb-07e7f620d0d2d3c30

Main network ACL
acl-0b778f9dc9df8f1a0

Default VPC
No

IPv4 CIDR
20.200.0.0/16

IPv6 pool
–

IPv6 CIDR (Network border group)
–

Network Address Usage metrics
Disabled

Route 53 Resolver DNS Firewall rule groups
–

Owner ID
112334179662

Encryption control ID
–

Encryption control mode
–

subnet-034fb45aa5df3ed75 / private subnet

Actions

Details

Subnet ID
subnet-034fb45aa5df3ed75

IPv4 CIDR
20.0.2.0/24

Availability Zone
aps2-az2 (ap-south-2b)

Default subnet
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:ap-south-2:112334179662:subnet/subnet-034fb45aa5df3ed75

Available IPv4 addresses
250

VPC
vpc-0121900efe558e38d | hyderabadvpc

Auto-assign public IPv4 address
No

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

Route table
rtb-027e0a7ae2f1f7efe | hypeering

Auto-assign IPv6 address
No

Hostname type
IP name

Owner
112334179662

Block Public Access
Off

IPv6 CIDR association ID
–

Network ACL
acl-0162ee07865da9c49

IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

subnet-0825199782687d450 / public subnet

Actions

Details

Subnet ID
subnet-0825199782687d450

IPv4 CIDR
20.0.1.0/24

Availability Zone
aps2-az1 (ap-south-2a)

Default subnet
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:ap-south-2:112334179662:subnet/subnet-0825199782687d450

Available IPv4 addresses
250

VPC
vpc-0121900efe558e38d | hyderabadvpc

Auto-assign public IPv4 address
Yes

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

Route table
rtb-0fd2d5c97bca0766e | hy route

Auto-assign IPv6 address
No

Hostname type
IP name

Owner
112334179662

Block Public Access
Off

IPv6 CIDR association ID
–

Network ACL
acl-0162ee07865da9c49

IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

VPC dashboard

Filter by VPC
Choose a VPC

AWS Global View

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

subnet-09a33777f458afd1a / privatesubnet

Actions

Details

Subnet ID
subnet-09a33777f458afd1a

IPv4 CIDR
20.200.1.0/24

Availability Zone
apse1-az2 (ap-southeast-1a)

Network ACL
acl-0b778f9dc9df8f1a0

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
–

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:ap-southeast-1:112334179662:subnet/subnet-09a33777f458afd1a

Available IPv4 addresses
250

Network border group
ap-southeast-1

Default subnet
No

Customer-owned IPv4 pool
–

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
–

VPC
vpc-0a5283e29e2d2a530 | sing-vpc

Auto-assign public IPv4 address
No

Outpost ID
–

Hostname type
IP name

Owner
112334179662

Block Public Access
Off

IPv6 CIDR association ID
–

Route table
rtb-012cee56a02ea1a35 | routesing

Auto-assign IPv6 address
No

IPv4 CIDR reservations
–

Resource name DNS A record
Disabled

rtb-027e0a7ae2f1f7efe / hypeering

Actions

Details Info

Route table ID
rtb-027e0a7ae2f1f7efe

VPC
vpc-0121900efe558e38d | hyderabadvpc

Main
No

Owner ID
112334179662

Explicit subnet associations
subnet-034fb45aa5df3ed75 / private subnet

Edge associations
–

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	pcx-00191c75f2a8d35b9	Blackhole	No	Create Route
20.0.0.0/16	local	Active	No	Create Route Table

rtb-0fd2d5c97bca0766e / hy route

Actions

Details Info

Route table ID
rtb-0fd2d5c97bca0766e

Main
No

Explicit subnet associations
subnet-0825199782687d450 / public subnet

Edge associations
–

VPC
vpc-0121900efe558e38d | hyderabadvpc

Owner ID
112334179662

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0cbd4ad6d33187d18	Active	No	Create Route
20.0.0.0/16	local	Active	No	Create Route Table

rtb-012cee56a02ea1a35 / routesing

Actions

Details Info

Route table ID
rtb-012cee56a02ea1a35

VPC
vpc-0a5283e29e2d2a530 | sing-vpc

Main
No

Owner ID
112334179662

Explicit subnet associations
subnet-09a33777f458afd1a / privatesubnet

Edge associations
–

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

IP version: Both

Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	pcx-00191c75f2a8d35b9	Active	No	Create Route
20.200.0.0/16	local	Active	No	Create Route Table

Instance summary for i-003a8d8f5668c7248 (ewb-bast) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-003a8d8f5668c7248

IPv6 address
–

Hostname type
IP name: ip-20-0-1-108.ap-south-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-0121900efe558e38d (hyderabadvpc)

Subnet ID
subnet-0825199782687d450 (public subnet)

Instance ARN
arn:aws:ec2:ap-south-2:112334179662:instance/i-003a8d8f5668c7248

Public IPv4 address
40.192.1.72 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-20-0-1-108.ap-south-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
20.0.1.108

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
40.192.1.72 [Public IP]

IAM role
–

IMDSv2
Required

Operator
–

Instance summary for i-09d676f4700768801 (ewb-app-a) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-09d676f4700768801

IPv6 address
–

Hostname type
IP name: ip-20-0-2-114.ap-south-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-0121900efe558e38d (hyderabadvpc)

Subnet ID
subnet-034fb45aa5df3ed75 (private subnet)

Instance ARN
arn:aws:ec2:ap-south-2:112334179662:instance/i-09d676f4700768801

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-20-0-2-114.ap-south-2.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
20.0.2.114

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

IAM role
–

IMDSv2
Required

Operator
–

Instance summary for i-071f78336757f19f5 (ewb-app-b) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID
i-071f78336757f19f5

IPv6 address
–

Hostname type
IP name: ip-20-200-1-203.ap-southeast-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0a5283e29e2d2a530 (sing-vpc)

Subnet ID
subnet-09a33777f458afd1a (privatesubnet)

Instance ARN
arn:aws:ec2:ap-southeast-1:112334179662:instance/i-071f78336757f19f5

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-20-200-1-203.ap-southeast-1.compute.internal

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more

Auto Scaling group name
–

Managed
false

Private IPv4 addresses
20.200.1.203

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

IAM role
–

IMDSv2
Required

Operator
–

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source type Info	Source Info
sgr-092e44b33d6fe369d	SSH	TCP	22	Custom	0.0.0.0/0

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source type Info	Source Info	Description - optional Info
sgr-04b10c0c2dc77dff9	MYSQL/Aurora	TCP	3306	Custom	20.0.1.0/24	
sgr-0636be5ac2f1180a6	SSH	TCP	22	Custom	20.0.1.108/32	
sgr-05119e7c35972de21	Custom ICMP - IPv4	Echo ...	N/A	Custom	20.200.1.203/32	

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source type Info	Source Info	Description - optional Info
sgr-0caf0dbb0d14abcde	MYSQL/Aurora	TCP	3306	Custom	20.0.2.0/24	
sgr-0630c12805bce199b	SSH	TCP	22	Custom	20.0.2.114/32	
sgr-077cfecb146a586b7	Custom ICMP - IPv4	Echo ...	N/A	Custom	20.0.2.114/32	

```
[ec2-user@ip-20-0-1-108 ~]$ ssh -i "ewb-app-a.pem" ec2-user@20.0.2.114
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 26 13:05:31 2026 from 20.0.1.108
[ec2-user@ip-20-0-2-114 ~]$ ssh -i "ewb-app-b.pem" ec2-user@20.200.1.203
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 26 13:12:15 2026 from 20.0.2.114
```

```
[ec2-user@ip-20-200-1-203 ~]$ ping -c 4 20.0.2.114
PING 20.0.2.114 (20.0.2.114) 56(84) bytes of data.
64 bytes from 20.0.2.114: icmp_seq=1 ttl=127 time=42.7 ms
64 bytes from 20.0.2.114: icmp_seq=2 ttl=127 time=42.7 ms
64 bytes from 20.0.2.114: icmp_seq=3 ttl=127 time=42.7 ms
64 bytes from 20.0.2.114: icmp_seq=4 ttl=127 time=42.7 ms

--- 20.0.2.114 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
```

```
[ec2-user@ip-20-200-1-203 ~]$ ping -c 4 20.0.2.114
PING 20.0.2.114 (20.0.2.114) 56(84) bytes of data.

--- 20.0.2.114 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
```

```
[ec2-user@ip-20-200-1-203 ~]$ ping -c 4 20.0.2.114
PING 20.0.2.114 (20.0.2.114) 56(84) bytes of data.
64 bytes from 20.0.2.114: icmp_seq=1 ttl=127 time=43.3 ms
64 bytes from 20.0.2.114: icmp_seq=2 ttl=127 time=43.4 ms
64 bytes from 20.0.2.114: icmp_seq=3 ttl=127 time=43.4 ms
64 bytes from 20.0.2.114: icmp_seq=4 ttl=127 time=43.4 ms

--- 20.0.2.114 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
```

AWS VPC Peering Project – What I Learned ☺

1. VPC Peering Concept

- VPC Peering allows two VPCs (in same or different regions) to communicate privately using the AWS network (not the internet).
- It is a point-to-point connection between VPCs.
- No transitive routing – VPC A cannot reach VPC C through VPC B.
- Both sides need proper route table entries and security group rules.

2. AWS Services & Resources Used

- VPC (Hyderabad & Singapore)
- Subnets (public & private)
- EC2 instances (Bastion, App-A, App-B)
- Internet Gateway (for bastion public access)
- Route Tables
- Security Groups
- VPC Peering Connection

3. Key Configurations

- Created 2 VPCs with different CIDRs:
  - Hyderabad VPC – 20.0.0.0/16
  - Singapore VPC – 20.200.0.0/16
- Added peering connection (pcx-00191c75f2a8d35b9) and verified it is ACTIVE.
- Added route table entries in both VPCs for the other VPC CIDR via the peering connection.
- Configured security groups to allow SSH and ICMP between instances.
- Ensured App-A and App-B do not have public IPs (private connectivity only).

* Key Takeaways

→ VPC peering = private, secure, direct connectivity.
→ Both VPCs need routes and SG rules.
→ No public IPs for private instances.
→ Always verify in both directions.
→ Proper documentation is essential for the project.

4. Connectivity Testing

- Bastion → App-A : SSH successful.
- App-A → App-B : Ping successful (0% loss).
- App-B → App-A : Ping successful (0% loss).
- App-B SSH : Connected successfully.
- Controlled failure : Removed ICMP rule from App-B SG → Ping failed (100% loss).
- Restored rule → Ping successful again.

5. Useful Commands

- ssh -i key.pem ec2-user@ <public-ip> (Bastion)
- ssh -i key.pem ec2-user@ <private-ip> (App-A/App-B)
- ping -c 4 <private-ip>
- ip route
- hostname
- hostname -I

6. Important AWS Console Checks

- VPC peering status (Active)
- Route tables (both sides have correct routes)
- Subnet associations
- Security group inbound rules
- Instances have no public IP
- Internet Gateway attached only to bastion VPC

7. What I Learned Overall

- How VPC peering works and why route tables and security groups are important.
- Difference between public and private subnets.
- How to troubleshoot connectivity issues.
- Importance of verifying both directions.
- How to document and present AWS project evidence.
- The value of step-by-step planning and checking.

Small steps... Big learning! ☺

Successfully initiated termination (deletion) of i-031f9c662b5d63442

Instances (1/1) Info

Last updated less than a minute ago

Connect | Instance state | Actions | Launch instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

1

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
ewb-app-b	i-031f9c662b5d63442	Running	t3.micro	3/3 checks passec	ap-southeast-1a	ec2-54-169-109-30.ap-...

You have successfully deleted peering connection pcx-00191c75f2a8d35b9.

1 of 1 related routes were deleted

Peering connections (1) Info

Actions ▼ | Create peering connection

Find peering connections by attribute or tag

1

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
singpeering	pcx-00191c75f2a8d35b9	Deleting	vpc-0a5283e29e2d2a530 / sing...	vpc-0121900efe558e38d

You successfully deleted vpc-0a5283e29e2d2a530 / sing-vpc and 1 other resources.

▶ Details

Your VPCs

VPCs | VPC encryption controls

Your VPCs (1) Info

Last updated less than a minute ago | Actions | Create VPC

Filter VPCs by attribute or tag

1

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	II
–	vpc-051c70edc0603bf03	Available	–	–	Off	1

Detaching internet gateway unsuccessful

- igw-0ec0b7c4353e5540a

Network vpc-051c70edc0603bf03 has some mapped public address(es). Please unmap those public address(es) before detaching the gateway.

Internet gateways (1/1) Info

Actions | Create internet gateway

Find internet gateways by attribute or tag

1

Name	Internet gateway ID	State	VPC ID	Owner
–	igw-0ec0b7c4353e5540a	Attached	vpc-051c70edc0603bf03	112334179662

Successfully initiated termination (deletion) of i-09d676f4700768801,i-003a8d8f5668c7248

Instances (2/2) Info

Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

1

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
ewb-app-a	i-09d676f4700768801	Shutting-d…	t3.micro	–	ap-south-2b	–
ewb-bast	i-003a8d8f5668c7248	Shutting-d	t3.micro	–	ap-south-2a	–

Detaching internet gateway unsuccessful

- igw-0cbd4ad6d33187d18 / hy-igw

Network vpc-0121900efe558e38d has some mapped public address(es). Please unmap those public address(es) before detaching the gateway.

Internet gateways (1/2) Info

Actions ▼ | Create internet gateway

Find internet gateways by attribute or tag

1

Name	Internet gateway ID	State	VPC ID	Owner
–	igw-09da7336a9e3d7e8a	Attached	vpc-066f9dd11dd564da7	112334179662
hy-igw	igw-0cbd4ad6d33187d18	Attached	vpc-0121900efe558e38d | hyderabadvpc	112334179662